

Data Flow Diagrams and User Stories

Phase 2: Requirement Analysis

shopEZ E-commerce Marketplace | MERN Stack | June 2026

1. Data Flow Diagrams (DFD) Overview

Data Flow Diagrams document how information moves through the shopEZ system across its three primary external entities: Customers, Administrators/Sellers, and integrated third-party services.

2. Level 0 DFD: Context Diagram

Customer (End User)

Inputs: Credentials, Cart Actions, Reviews | ^ Outputs: Catalog, Confirmations

shopEZ E-Commerce System

Inputs: Product CRUD, Coupon Rules | ^ Outputs: Sales Analytics, Order Logs

Admin / Seller (Operations Manager)

3. Level 1 DFD: Process Breakdown

Process 1: User Authentication Flow

Step

Action

1	Customer inputs email and password into React Login form.
2	Axios POST request sent to /api/users/login on Express server.
3	authMiddleware.js validates credentials against User model in MongoDB.
4	Server signs and returns JWT token containing user._id and user.role.
5	Frontend stores token in HTTP-Only cookie; user state updated in AuthContext.

Process 2: Product Browsing Flow

Step	Action
1	React catalog triggers GET /api/products?search=... request.
2	productController.js constructs a Mongoose query with \$regex, \$gte, and \$lte filters.
3	Database executes the filtered text search and returns matching product documents.
4	Server serializes and returns filtered product list as JSON response.
5	React renders product grid cards with real-time filter and sort options applied.

Process 3: Cart Checkout Flow

Step	Action
1	Customer clicks Place Order; React sends POST /api/payment/create-intent with cart total.
2	Mock payment endpoint returns simulated client_secret string.

3	React confirms payment, then sends POST /api/orders with full cart, address, and coupon data.
4	orderController.js saves the Order document to MongoDB and decrements Product stock counts atomically.
5	Order confirmation page loads with itemized receipt and fulfillment tracking timeline.

Process 4: Admin Order Fulfillment

Step	Action
1	Admin logs in and accesses the Orders panel in AdminDashboard.jsx.
2	Admin selects an order and clicks the Advance Status button.
3	Axios PUT /api/orders/:id/deliver sends the new status milestone to the backend.
4	orderController.js updates the Order document's tracking timeline array in MongoDB.
5	Customer's MyOrders.jsx page reflects the updated stepper position on next load.

4. User Story Mapping

User Goal	Activity	Mapped USN
Register and Log In	Account creation, authentication, session management	USN-001, USN-002, USN-003
Browse and Discover	Catalog search, filter, sort, product detail view	USN-004
Save and Cart Items	Add to cart, manage wishlist, move to cart	USN-005, USN-006

Purchase	Checkout wizard, coupon apply, mock Stripe payment	USN-007, USN-008
Track and Review	Order history, fulfillment timeline, submit review, helpful vote	USN-009, USN-010
Administer	Product CRUD, user management, coupon management, analytics	USN-011